CHAPTER 14 · DERIVATIONS

CAPACITORS

4 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 CAPACITANCE OF A PARALLEL PLATE CAPACITOR

SYLLABUS 14.1(D)

RESULT $C_0 = \dfrac{\varepsilon_0 A}{d}$

1. Plates of area A, separation d, charges ±Q: $\sigma = \dfrac{Q}{A}$.
2. Gauss's law: the field between the plates is $E = \dfrac{\sigma}{\varepsilon_0} = \dfrac{Q}{\varepsilon_0 A}$ (uniform).
3. $V = Ed = Qd/(\varepsilon_0 A)$.
4. $\mathbf{C_0 = \dfrac{Q}{V} = \dfrac{\varepsilon_0 A}{d}}$.

2 CAPACITORS IN SERIES AND IN PARALLEL

SYLLABUS 14.3(A)

RESULT series: $\dfrac{1}{C} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \ldots$; parallel: $C = C_1 + C_2 + \ldots$

1. **Series**: each capacitor carries the same charge Q, and $V = V_1 + V_2 + V_3$.
2. $\dfrac{Q}{C} = \dfrac{Q}{C_1} + \dfrac{Q}{C_2} + \dfrac{Q}{C_3}$, so $\mathbf{\dfrac{1}{C} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \dfrac{1}{C_3}}$.
3. **Parallel**: each capacitor has the same p.d. V, and $Q = Q_1 + Q_2 + Q_3$.
4. $CV = C_1V + C_2V + C_3V$, so $\mathbf{C = C_1 + C_2 + C_3}$.

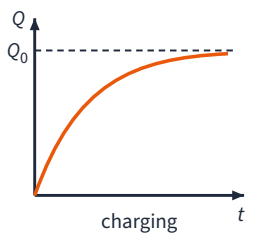
Energy formulae $U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \dfrac{Q^2}{2C}$: derivation not required (14.4).

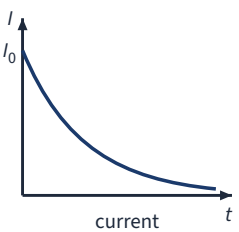
3 CHARGING THROUGH A RESISTOR

SYLLABUS 14.5(B)

RESULT $Q = Q_0(1 - e^{-t/RC})$, $V = V_0(1 - e^{-t/RC})$, $I = I_0 e^{-t/RC}$

1. Kirchhoff's law round the circuit (supply V_0): $V_0 = IR + \dfrac{Q}{C}$, with $I = \dfrac{dQ}{dt}$.
2. $R\dfrac{dQ}{dt} = \dfrac{CV_0 - Q}{C}$, so $\dfrac{dQ}{CV_0 - Q} = \dfrac{dt}{RC}$.
3. Integrate from Q = 0 at t = 0: $-\ln\dfrac{CV_0 - Q}{CV_0} = \dfrac{t}{RC}$.
4. $\mathbf{Q = Q_0(1 - e^{-t/RC})}$, with $Q_0 = CV_0$; dividing by C: $\mathbf{V = V_0(1 - e^{-t/RC})}$.
5. $I = \dfrac{dQ}{dt} = \dfrac{Q_0}{RC}e^{-t/RC}$: $\mathbf{I = I_0 e^{-t/RC}}$, with $I_0 = \dfrac{V_0}{R}$.





4 DISCHARGING THROUGH A RESISTOR

SYLLABUS 14.5(D)

RESULT $Q = Q_0e^{-t/RC}$, $V = V_0e^{-t/RC}$, $I = I_0e^{-t/RC}$

1. With the supply removed: $0 = IR + \frac{Q}{C}$, and $I = \frac{dQ}{dt}$.
2. $\frac{dQ}{dt} = -\frac{Q}{RC}$, so $\frac{dQ}{Q} = -\frac{dt}{RC}$.
3. Integrate from Q_0 at $t = 0$: $\ln\frac{Q}{Q_0} = -\frac{t}{RC}$, so $\mathbf{Q = Q_0e^{-t/RC}}$ and $\mathbf{V = V_0e^{-t/RC}}$.
4. Size of the current: $I = Q_0e^{-t/RC}/RC$, i.e. $\mathbf{I = I_0e^{-t/RC}}$ (in the opposite direction to the charging current).